MIL-T-60669(MU)
18 July 1968
SUPERSEDING
FA-PD-MI-2675
28 June 1965

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M19
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M19, Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F10520989	- Thruster, Cartridge Actuated, M19, Metal Parts Assembly
C8836128	- Container, Ammunition, Fiber M445, for Thruster, Cartridge Actuated, M19 or M13 with Cartridge M119

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

2.1 DRAWINGS (Cont'd)

Munitions Command

C8836129 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M19 or M13 with Cartridge M119 in Fiber Container M445

IEL-10520989 - Index to Inspection Equipment Lists for Thruster, Cartridge Actuated M19, Metal Parts Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Materials.- Components for remanufacture may be obtained from thruster models of the T-25 and M19 type.

3.3 Hydrostatic pressure.- The body shall withstand a constant internal pressure of 11,500 pounds per square inch (psi), minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.2).

3.4 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled with sufficient torque to lightly bottom components, except as noted in Table I.

3.4.1 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

3.4.1 (Cont'd)

TABLE I

Component	A Run On (minimum)	B Seating or Bottoming (inch pounds)	C Breakaway (inch pounds)
Piston and Piston Fastener	---	100 ± 20	75 min.

3.4.2 The remanufactured thruster metal parts assembly shall contain the following new components: Firing Pin Shear Pin, Piston Shear Pins (2), and O-rings.

3.4.3 The remanufactured thruster metal parts assembly shall be assembled with Breech, Part No. 10520977. The overall length (1.637 + .005) shall be checked to determine if components are like the noted Part No. All other breech components shall be discarded.

3.4.4 The remanufactured thruster metal parts assembly shall be assembled with the following component parts. Components that do not meet these requirements can be modified and used for assembly.

3.4.4.1 Base Cap, 10520982, covering a detail of drawing C10520975. The following dimensions shall be required for part to meet the latest drawing requirements: .568 - .015; .338 - .005 and .818 - .002 (See Figure I).

3.4.4.2 Piston Fastener, 10520983, covering a detail of drawing C10520980. The following dimension shall be required for part to meet latest drawing requirement: .487 + .010 (See Figure II).

3.4.4.3 Body, 10520990, covering a detail of drawing F10520989. The following dimension shall be required for part to meet latest drawing requirement: .887 + .002 (See Figure III).

3.5 Protective finish.- Protective finish shall be in accordance with the applicable specification and drawings and shall apply to all surfaces of the component regardless of contour configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (See 6.4).

3.6 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches (See 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, or other metal defects.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 and as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lots of metal parts assemblies shall consist of 1200 thrusters, maximum, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2 Examination.- The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be either reprocessed or rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2.2 (Cont'd)

TABLE III (Cont'd)

Sample Sizes for Minor Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.2.2.3 Ring, Drawing C10520489, covering a detail of Drawing F10520989

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Pin, Firing, Drawing C10520978, covering a detail of Drawing F10520989

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.2.2.5 Cap, Base, Drawing C10520982, covering a detail of Drawing F10520989	
CRITICAL	
None defined	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corners	Visual
105. Foreign matter	Visual
MINOR	
None defined	
4.2.2.6 Fastener, Piston, Drawing C10520983, covering a detail of Drawing F10520989	
CRITICAL	
None defined	
MAJOR	
101. Threads damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2.7 Body, Drawing F10520990, covering a detail of Drawing F10520989

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corners	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.8 Piston, Drawing D10520992, covering a detail of Drawing F10520989

CRITICAL	
None defined	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corners	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.9 Base Cap and Pellet Assembly, Drawing C10520975, covering a detail of Drawing F10520989

CRITICAL	
None defined	

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2.9 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pellet missing, loose or improperly inserted	Manual and Visual
102. Thread damaged	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.10 Piston and Pellet Assembly, Drawing C10520991, covering a detail of Drawing F10520989

CRITICAL	
None defined	
MAJOR	
101. Pellet missing, loose or improperly inserted	Manual and Visual
102. Thread damaged	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.11 Piston Fastener and Pellet Assembly, Drawing C10520980, covering a detail of Drawing F10520989

CRITICAL	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2.11 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pellet missing loose or improperly inserted	Manual and Visual
102. Thread damaged	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Breech and Firing Pin, Sub-assembly, consists of: Breech 10520977, Firing Pin 10520978 and O-Ring 501217, covering a detail of Drawing F10520989

CRITICAL	
1. Firing pin protrusion	Gage
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.13 Metal Parts Assembly, Drawing F10520989

CRITICAL	
None defined	
MAJOR	
101. Piston not free to rotate 360 degrees clockwise	Manual
102. Any part missing, except as otherwise classified	Visual
103. Improper assembly	Visual
104. Thread damaged or not full	Visual
105. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

a box type housing preserved as specified in 5.2. shall be packed i

4.2.2.13 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Protective finish inadequate or defective	Visual
202. Barrier material missing from set screws	Visual
203. Shipping cap or plug missing	Visual
204. Pellet missing from piston fastener	Visual

4.2.2.14 Fiber Container, Drawing C8836128

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.15 Box, Packing, Drawing C8836129

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

4.2.2.15 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Hydrostatic pressure.- Each body shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the unit.

4.2.3.2 Breakaway torque.- Each piston fastener and piston shall be subjected to a breakaway torque. Failure of any unit to comply with the requirements of Column C in 3.4.1 shall be cause for rejection of the unit.

4.2.3.3 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply shall be cause for rejection of the lot.

4.2.3.4 Residual magnetism.- Each assembly shall be subjected to a test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520989.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- The test shall be performed after application of protective finish. The testing device used shall internally seal the body at each end without applying any axial load other than that applied by the liquid pressure.

4.3.2 Breakaway torque.- A torque wrench set at the torque requirement specified in Table I shall be used in conjunction with a fixture to determine the adequacy of the joint. The piston shall be held secure and the torque shall be applied to the piston fastener.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

4.3.3 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.3.4 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8836128 and C8836129.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 When remanufacturing and loading of assembly is performed by the same facility, the requirement of 3.6 is waived and the residual magnetism performed on the loaded assembly only.

6.4 Special instructions covering the modification, repair and assembly of remanufactured components shall be as follows:

6.4.1 Protective finish.- Remanufactured components which include Base Cap, Body and Piston Fastener shall be stripped of protective finish and replated in accordance with applicable drawings and specifications.

6.4.2 Nylon pellets.- Base caps and piston fasteners which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed after replating in accordance with the applicable drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

6.4.3 Modification.

6.4.3.1 The component parts; Body, Base Cap and Piston Fastener, from T25 Thruster models only, will have to be modified in accordance with Figures I, II and III.

6.5 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2675, dated 28 June 1965.

CUSTODIAN:

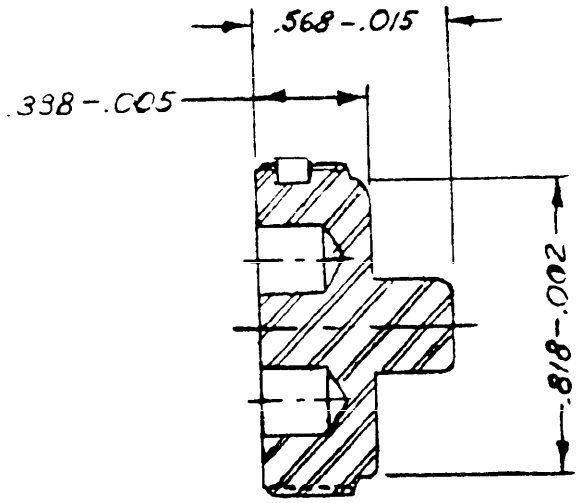
Army - MU

PREPARING ACTIVITY:

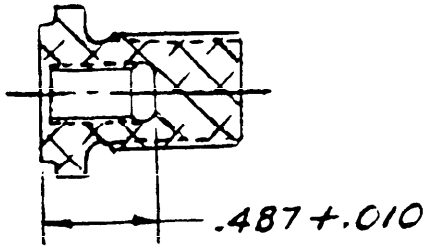
Army - MU

Project No. 1377-A170

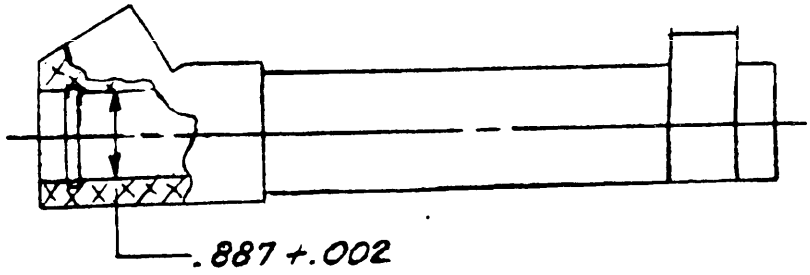
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.



BASE CAP
FIGURE I



PISTON FASTENER
FIGURE II



BODY
FIGURE III

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:54Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

FOLD

343-218 (A-105)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026\10-23T00:54Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.